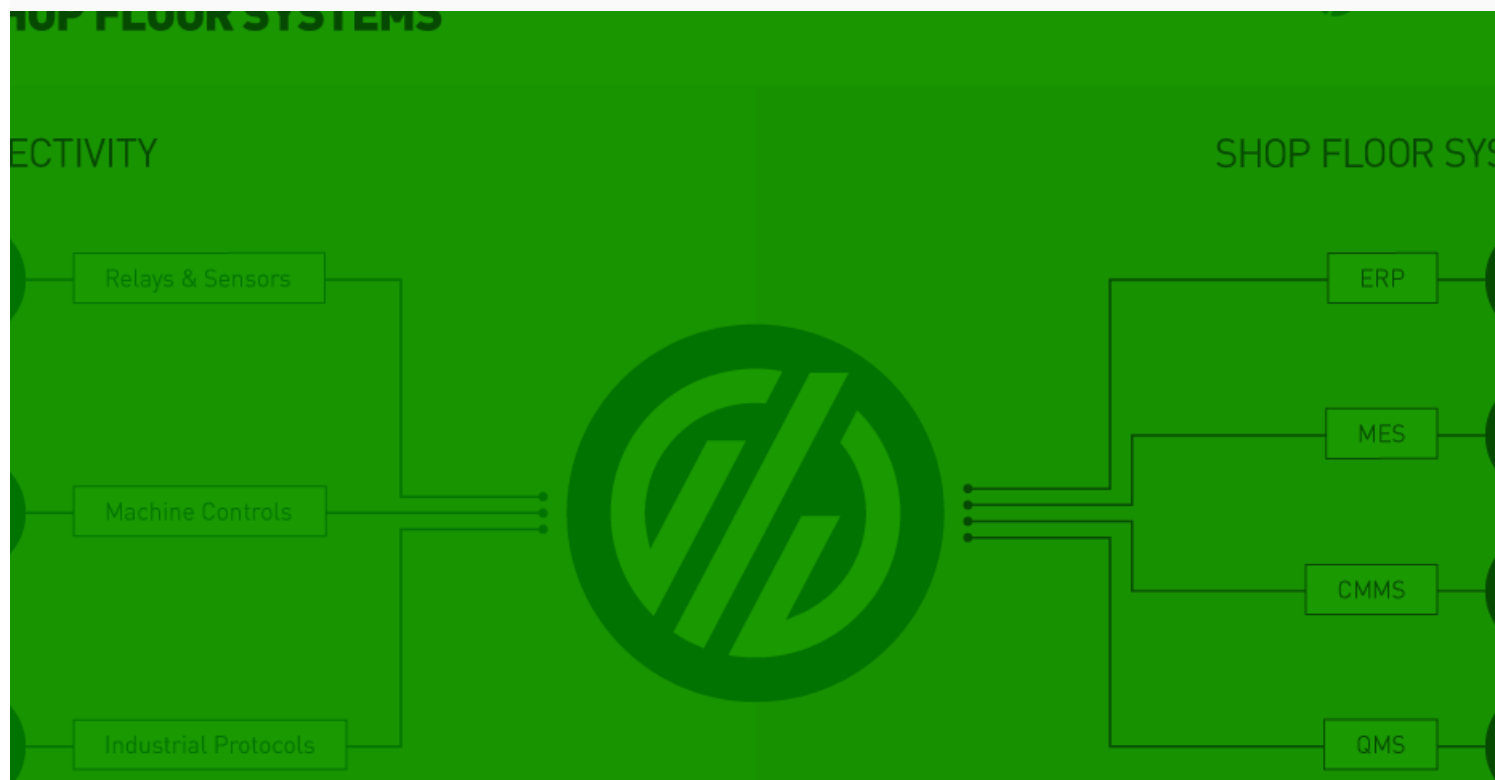


**HENRY FOSTER**

MachineMetrics, MES Software, MES, Death Of MES, Connected Factory / Last updated: September 24, 2025



MANUFACTURING EXECUTION SYSTEM (MES): A DEFINITIVE GUIDE

Manufacturing today is being reshaped by disruptive market trends and advancements in technology. Labor shortages, error-prone data, and clunky generalized legacy systems hold companies back at a time when they most need to be agile.

While Manufacturing Execution Systems (MES) have long been a part of the production environment, legacy MES has failed the modern manufacturer. These systems are unable to tie real-time shop floor data into the production management system, leading to a poor feedback loop.

By tying an MES to **automated, accurate data, companies can more effectively track downtime, eliminate process bottlenecks, and increase capacity.** They can also automate their OEE reporting, build dynamic and achievable schedules, and track work orders and machine performance without reliance on humans.

This is leading to the **next generation of MES systems.**





Understanding Manufacturing Execution Systems

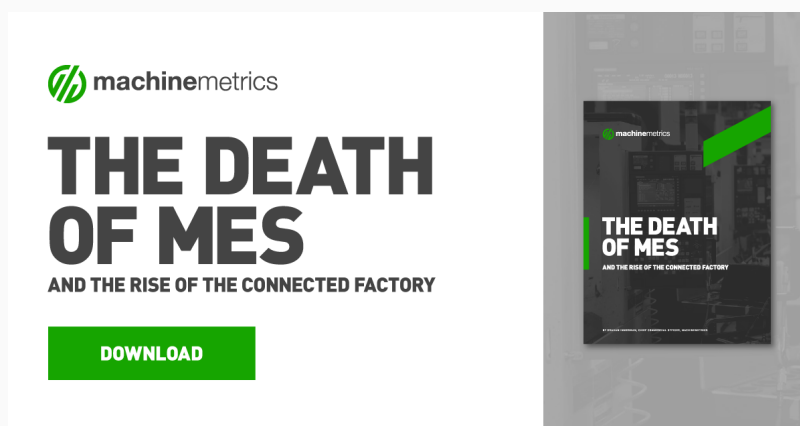
What is MES and Its “Core Four”?

A Manufacturing Execution System (MES) is a digital solution specifically designed to manage and monitor the production process in a manufacturing organization. Capturing real-time data from the factory floor offers insights into the entire manufacturing lifecycle, from raw materials to finished products.

MES is a powerful tool for manufacturers, enabling them to optimize their operations, enhance product quality, reduce waste, and ensure compliance with industry standards and regulations. It acts as the bridge between shop-floor systems and enterprise-level management, enabling a seamless flow of information and more informed decision-making.

MES was seen as a solution to some of the core production challenges associated with ERP systems. Although ERP has been seen as the source of truth and the core management solution for businesses, it exists far away from the shop floor. MES acted as a management extension of ERP, reaching closer to the shop floor activities and attempting to capture data from operations.

At its core, a manufacturing execution system offers four basic functions, commonly referred to as the “Core Four”: work order tracking, scheduling, measuring overall equipment effectiveness (OEE), and tracking downtime. Additionally, it offers various features that companies customize to meet their specific needs. These foundational elements constitute the central framework upon which any MES is built, but there is ofcourse many other modules within MES solutions.



What are the Challenges and Considerations of MES?

MES can be a great step in the right direction, but there are a number of items to consider when it comes to standard legacy MES solutions:

Markets Move Faster than Rigid Solutions:

Manufacturers face the challenge of keeping up with rapidly moving markets and competitive pressures. While there is much hype around IoT and digital transformation, manufacturers are well aware of their need to adapt to data-driven methodologies. Traditional MES and ERP systems often trap manufacturers with error-prone data, rigid processes, and delayed feedback loops, making it difficult to adapt quickly.

Blanket Solutions for a Diversified Industry

The manufacturing industry has become highly specialized, with diverse needs, processes, equipment, and performance indicators. Especially in discrete manufacturing, there is an incredible amount of variability within organizations. Legacy MES solutions are often broad, incomplete, and inflexible, forcing manufacturers to adapt their processes to the system rather than the other way around. Customization options are expensive, difficult to maintain, and challenging to scale, stifling innovation and continuous improvement initiatives.

Reliance on Humans Amidst a Labor Crisis:

MaxAI has arrived! Learn all about it [here](#)



Platform Solutions Company

Resources Why MM?

BOOK A DEMO
Sign In

SEE IT IN ACTION

CONTENTS

Understanding
Manufacturing
Execution
Systems

What are the
Challenges and
Considerations
of MES?

Key
Components of
MES

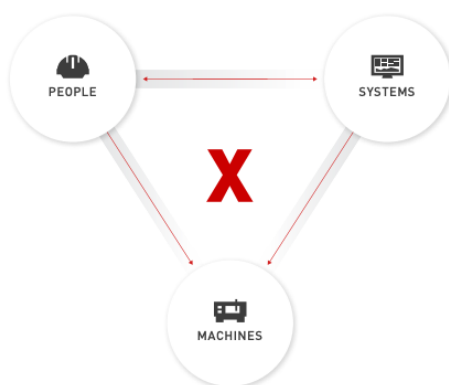
Advantages of
MES

Implementation
of MES in
Manufacturing

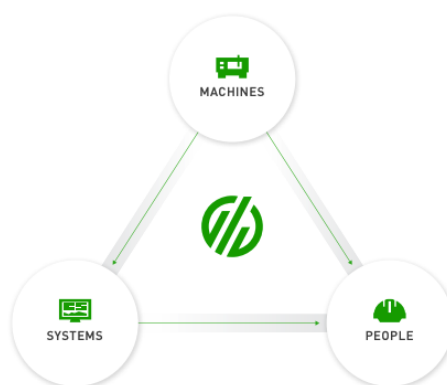
Integration with
Industry 4.0
Technologies

Final Thoughts

PEOPLE-DRIVEN STRATEGY



MACHINE-DRIVEN STRATEGY



Key Components of MES

The critical components of MES reflect the underlying functionalities that optimize the production process. Although this is not an exhaustive list, it highlights some of its significant components:

- Production and downtime tracking components follow the progress of goods through all stages of production, including timestamps and quantities.
- Quality management refers to tools that ensure products are manufactured to predefined standards and comply with regulations, including defect tracking and resolution management.
- Performance analysis is accomplished using analytical tools and reporting features that evaluate production data to gauge efficiency and productivity. This feature can help with planning and continuous improvement.
- Resource scheduling and dispatching manage and optimize the allocation of labor, materials, and equipment to meet production schedules.

Advantages of MES

MES is a transformative force in today's fast-paced industrial environment, reshaping how businesses monitor, control, and execute their operations.

businesses monitor, control, and execute their operations.

By integrating MES into manufacturing processes, businesses can gain a strategic advantage. MES enables accurate tracking of production activities, enhances efficiency, and drives quality improvements. Let's examine these benefits individually, focusing on productivity, quality control, and communication.

Increased Efficiency and Productivity

MES significantly increases efficiency and productivity by streamlining the information flow across the production cycle, reducing cycle times through better scheduling, and aligning the workforce with the most critical production tasks, optimizing labor utilization and overall productivity.

MES can also optimize machine schedules by monitoring machine performance, identifying bottlenecks in the manufacturing process, reducing idle times, and improving overall equipment effectiveness (OEE).

Enhanced Quality Control and Compliance

Quality control and compliance are among the top benefits of MES that significantly contribute to the success and reliability of manufacturing businesses. Here is how they benefit users of MES:

MES systems ensure that all processes adhere to predefined quality standards. By monitoring production in real time, MES can detect deviations from quality parameters, allowing immediate corrective action to maintain consistent product quality.

The system provides detailed tracking of materials and components throughout the production process, essential for determining the origin and history of finished products.

The immediate feedback on production variances provided by MES systems empowers users to adjust processes before they result in a non-conforming product. This proactive approach minimizes waste and significantly reduces the cost associated with rework and scrap, leading to substantial cost savings.

Data Collection and Analysis

While the above advantages are achievable, they require manufacturers to embrace the missing piece of the puzzle. Because traditional MES systems are overly reliant upon people, it still retains elements of manual data tracking and process monitoring.

MachineMetrics allows you to unlock shop-floor data to **tie real-time data acquisition and analysis to every machine, part, and work order**. This creates capabilities that are data-driven, accurate, extensible, and relevant to the unique production needs of each company.

Without a solution like MachineMetrics, the MES will likely be overly reliant on people, reducing its effectiveness.

Implementation of MES in Manufacturing

Implementing MES involves a multi-step process that requires thorough planning, careful execution, and ongoing support. Initially, companies must define their objectives for MES, such as improving product quality, increasing efficiency, or gaining better visibility into production processes.

Following software selection, a team member should develop a detailed project plan outlining the implementation phases, timelines, and responsible parties. Users training is essential to ensure they are proficient in using the MES and can leverage its full capabilities. Finally, continuous system monitoring and optimization are vital for promptly addressing issues and adapting to evolving manufacturing needs.

One of the core challenges with MES is that the customization needed to fit a given organization generally results in extensive costs and time for implementation. Further, this leads to issues down the road, as any updates and changes will require additional engineering costs.

Integration with Industry 4.0 Technologies

MES is at the forefront of integrating with Industry 4.0 technologies, such as the Internet of Things, to revolutionize the manufacturing industry. By harnessing IoT, MES can collect real-time data from machines and equipment across the factory floor, enabling precise monitoring and control of manufacturing processes.

Want to see what this looks like? Watch a live demo:



This connectivity ensures that decision-makers have access to up-to-date information, facilitating rapid responses to any arising issues. Simultaneously, AI complements this by analyzing the vast amounts of data generated, identifying patterns, and predicting potential faults before they occur.

Final Thoughts

Is MES the answer to your problems? While MES offers an opportunity to bring the management help of ERP closer to shop floor operations, existing systems are failing to drive the promised value.

With massive customization required and a lack of accurate, real-time shop floor data, MES does not drive the value and promise of digital transformation.

Manufacturers should consider leveraging point-based solutions to solve their critical needs and functions instead of relying on one system to try to "do it all".

**RT DRIVING DECISIONS
H MACHINE DATA.**

READY TO EMPOWER YOUR SHOP FLOOR?

[Learn More →](#)

COMMENTS

LEAVE A COMMENT

First Name*

Last Name

Email*

Website

Comment*

protected by reCAPTCHA



SUBMIT COMMENT

SUBSCRIBE TO OUR MAILING LIST

Email*

SUBSCRIBE

RELATED POSTS

READ THE LATEST



READ PRESS RELEASE

LAUNCHES - CLOSING...

On May 16, 2025 – MachineMetrics, the leading MES and industry solution, today announced the launch of a powerful solution that streamlines planning and scheduling for manufacturing, unlocking complete



THE CHALLENGE OF TRACKING MANUAL PROCESSES IN...

The Hidden Half of Manufacturing Despite massive investments in automation, the majority of manufacturing work is still performed by people. In fact, studies show that 72% of factory tasks are carried out manually - across assembly, inspection, finishing, material handling, and countless other



MACHINOMETRICS, PARTNER TO DELIV

EASTHAMPTON, MA – September 16, 2025 – MachineMetrics, the leading MES and industry solution, today announced the launch of a powerful solution that streamlines planning and scheduling for manufacturing, unlocking complete

of the production routing. nation efforts have pment, the majority of...

ricsproduct-Updates /

operations. Although there are expectations that machines and robot will eventually overtake many of these tasks, it's clear that in the short...

Machinemetricsmes / June 11, 2024

machines break down, set arrive late, and tribal know logic. This is the execution

Press-Releasemachineme Partnerships / June 11, 20:

SEE ALL POSTS

POPULAR

8 WASTES OF LEAN MANUFACTURING

MachineMetrics / Jan 25, 2016

QUALITY CONTROL IN MANUFACTURING: USING DATA TO IMPROVE YOUR QUALITY PROGRAM

MachineMetrics Quality Control / Oct 27, 2022

TAKT TIME VS CYCLE TIME VS LEAD TIME | DEFINITIONS AND CALCULATIONS

MachineMetrics manufacturing analytics / Sep 24, 2020

FIRST PASS YIELD: WHAT IS IT, FORMULA, AND HOW TO IMPROVE

MachineMetrics Quality Assurance / Sep 22, 2021

A GUIDE TO CUTTER COMPENSATION: WHAT IS G41 AND G42?

MachineMetrics Tool Monitoring / Feb 03, 2022

CATEGORIES

MACHINEMETRICS (422)

LEAN MANUFACTURING (74)

INDUSTRY 4.0 (50)

PRODUCT UPDATES (25)

MANUFACTURING DATA (23)

MANUFACTURING NEWS (20)

MES (19)

DATA SCIENCE (15)

OEE (14)

DATA COLLECTION (14)

[All Categories +](#)

Contact Us

844-822-0664

info@machinemetrics.com

Easthampton Office

116 Pleasant St, Suite
316, Easthampton, MA
01027

Platform

Platform Overview

Intelligent MES

Machine Monitoring

Edge Platform

Cloud Platform

APIs & Applications

Industrial IoT Platform

Applications

Production Monitoring

Condition Monitoring

Predictive Maintenance

Process Optimization

For Machine Builders and Distributors

For Developers

Production Schedule Intelligence

Job Tracking

Industry Solutions

Aerospace & Defense

Automotive

Contract Manufacturers

Heavy Machinery

Medical devices

Oil & Gas

Machine Monitoring

Precision Metalworking

Swiss CNC

ERP Integration

Metal Stamping & Fabrication

Tool, Die & Mold CNC

Resources

Blog

ROI Calculator

Waste Calculator

Downloads

Newsroom

Connectivity Hub

Sitemap

Company

Team

Careers

Events

Partner Program

Support

Privacy Policy

Data Processing Addendum

Service Level Agreement

Return Policy

Website Terms

Contact Us

Sign In

machinemetrics

All Rights Reserved by MachineMetrics ©2026

https://www.machinemetrics.com/blog/manufacturing-execution-system-mes?utm_source=chatgpt.com

Page 8 of 8